



Sri Sivasubramaniya Nadar College of Engineering, Kalavakkam – 603 110
(An Autonomous Institution, Affiliated to Anna University, Chennai)

Department of CSE
Continuous Assessment Test – II
Regulations – R2021

Degree & Branch	BE CSE				Semester	V
Subject Code & Name	UCS 2501 Computer Networks					
Academic Year	2023-2024 ODD	Batch	2021-2025	Date	02.11.2023	FN
Time: 08:10 - 09:40 a.m (90 Minutes)	Answer All Questions				Maximum: 50 Marks	

(K1: Remembering, K2: Understanding, K3: Applying, K4: Analyzing, K5: Evaluating)

CO1:	Describe the principles, design, terminology, and concepts of the network models (K2)
CO2:	Select the protocols of data link layer and apply error control mechanisms (K3)
CO3:	Choose the core functions of network layer protocols and apply them for data communication (K3)
CO4:	Select the Transport Layer Protocols suitable for transmission of data. (K3)
CO5:	Illustrate the Application Layer Protocols and their basic functionalities (K2).

Part – A ($4 \times 2 = 8$ Marks)

		KL	CO	PI
1.	A client uses UDP to send data to a server. The data length is 16 bytes. Make use of UDP and calculate the efficiency of this transmission at the UDP level (ratio of useful bytes to total bytes).	K3	CO3	1.1.2, 2.1.3, 13.3.1
2.	Mention the port numbers of different connections used in FTP and their usages.	K1	CO5	1.3.1, 1.4.1, 2.1.2, 13.1.1
3.	Define probe. What is the threshold value of the persistence timer?	K1	CO3	1.3.1, 1.4.1, 2.1.2, 13.1.1
4.	Apply the error control of TCP, state the rule along with its number that can be applied for this scenario.	K3	CO4	1.3.1, 2.3.1, 13.3.1

10.	Apply TCP Reno algorithm to plot the congestion window graph versus RTT. The data transmission starts with $cwnd=1$, $ssthresh=16$ at $RTT=0$. At $RTT=1$, $cwnd=2$ and so on following TCP Reno principle. The following sequence of events occurs, 1. At $RTT=3$, the timeout occurs. 2. At $RTT=13$, three dupACK occurs. 3. At $RTT=16$, new ACK arrives. 4. Stop the graph at $RTT=20$ Draw the sender side FSM of TCP Reno.	K3	CO4	2.1.2, 2.1.3, 2.3.1, 13.3.1
(Or)				
11.	Apply the Go Back N and Selective Repeat flow control protocols for the following scenario and illustrate the flow diagram for the same. Packets 0, 1, 2, 3 and 4 are sent. However, packet 2 is lost. The receiver receives packets 3 and 4. Complete this sequence using both the protocols. Indicate the pointers. The maximum window size is 8.	K3	CO4	2.1.2, 2.1.3, 2.3.1, 13.3.1